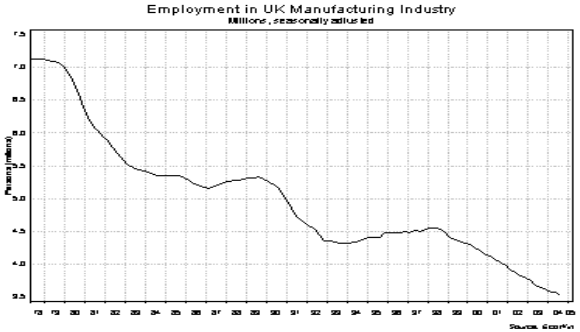
CPI – UK Graphene Application Centre

CPI Approach to Innovation, to Realise Commercial Value from Graphene

IDTechEx – Santa Clara

Tom Taylor – 19 November 2015

Manufacturing in the UK 1750 to 1980



However

Many Innovative UK SME’s replacing traditional manufacturing Industry with high value future applications, such as pragmatIC

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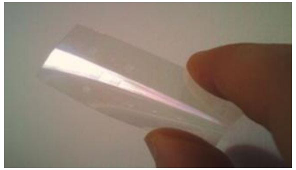
PragmatIC Printing enables
“Printed Logic”: Integrated Circuits (ICs) on plastic
that introduce electronic intelligence and interactivity
into a wide range of novel product form factors:
thin, flexible, transparent, robust, disposable, …



Printed Electronics manufacturing in Healthcare

- Global healthcare spend is $8.5 Trillion of which $450B (~5%) is medical technology
- 70% - 80% of healthcare spend
  - Asthma
  - Chronic Obstructive Pulmonary Disease
  - Diabetes
  - 300M people in both the EU and USA have at least of these conditions
- "Aging Society" and tele-Health or mobile health : $9.8B in 2010

Polyphotonix sleep mask

- Treatment for diabetic retinopathy
- An eye disease that leads to blindness in diabetics due to poor blood circulation
- Conventional treatments are expensive, invasive and in the case of laser ablation, lead to loss of sight
- Sleep mask stimulates the eye at night, so encouraging greater circulation of blood oxygen
- Estimated to save UK Health service £1bn per year
- Trial in EU countries
- Developed and manufactured in the UK

Polyphotonix sleep mask

UK is a great place to manufacture

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Funding

UK TSB – Technology strategy board – Technology Inspired Research £200m pa

EU Horizon 2020 (E70bn), - with £17bn for Industrial Innovation

Scientific Leadership

Centres for Innovative Manufacturing – research council funded

Some of our best scientific and engineering minds applied to manufacturing methods

Ultra-Precision Engineering – Cambridge and Cranfield

CIM in Large Area Electronics – Cambridge, Imperial col, Manchester, Swansea

Scale-up facilities

UK Catapult Centres

Seven National funded centres for scale up and exploitation – open to all

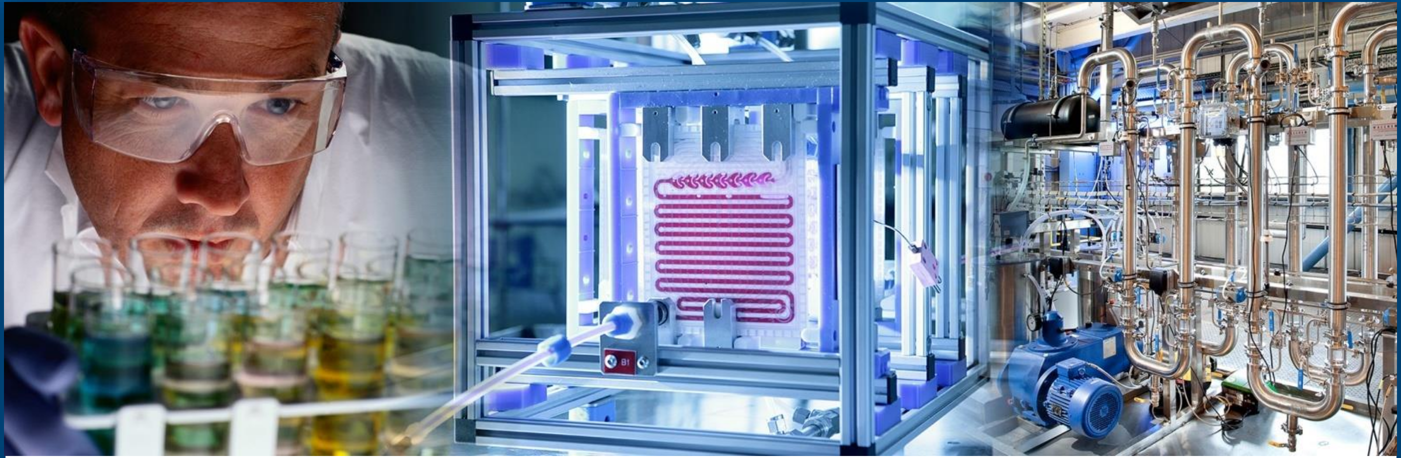
CPI – electronics, MTC for robotics, NCC for composites integration

CPI Printable Electronics Centre introduction

- UK national facilities for
  printed and plastic electronics, formulation and graphne
- Purpose - scale up from R&D to pilot processing
- 1500 m^2 of clean room facilities

Agenda – Graphene Application Innovation centre

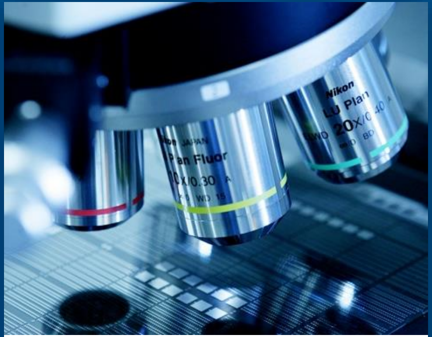
- Strategic context – Printable Electronics and Formulation
- Our offer – Innovation and innovation integrator
- Risks of R&D and scale up
- Engagement
- Equipment
- Progress to date: build, market, projects, Manchester, Cambridge,



CPI: From Innovation to Commercialisation

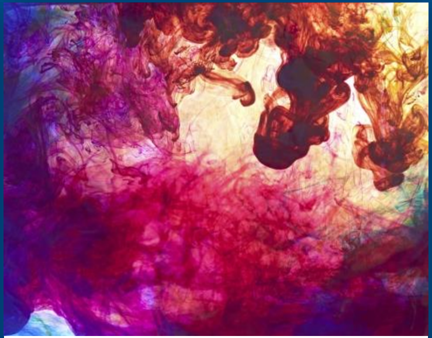
Technology Focus

Industrial Biotechnology and Biorefining



Printable Electronics

Biologics



Formulation and Flexible Manufacturing

CPI works across a distinct set of technological areas that offer the largest potential impact on the future of UK and global manufacturing.

Who are CPI?

CPI is a UK technology innovation centre and the process element of the Government's High Value Manufacturing Catapult.

National Printable Electronics Centre

Sustainable Processing Centre

Formulation and Flexible Manufacturing

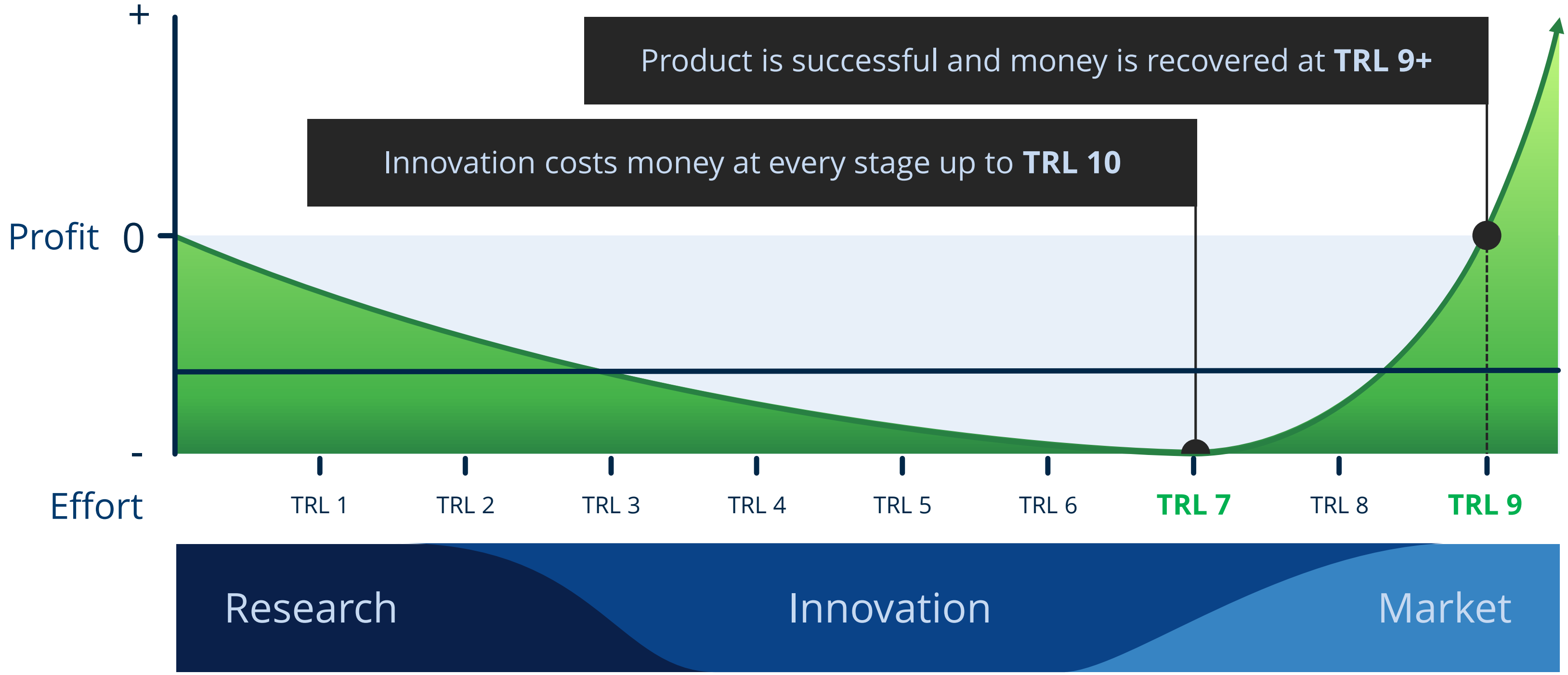
National Biologics Manufacturing Centre

We use applied knowledge in science and engineering combined with state of the art facilities to enable our clients to **develop, prove, prototype** and **scale-up** the next generation of products and processes.

What is Innovation?

The art or science of converting **existing inventions** and **ideas** into **practical products** or **services** that can be used in everyday life

Where does Innovation fit in?



The CPI Innovation Integrator^{TM} Process

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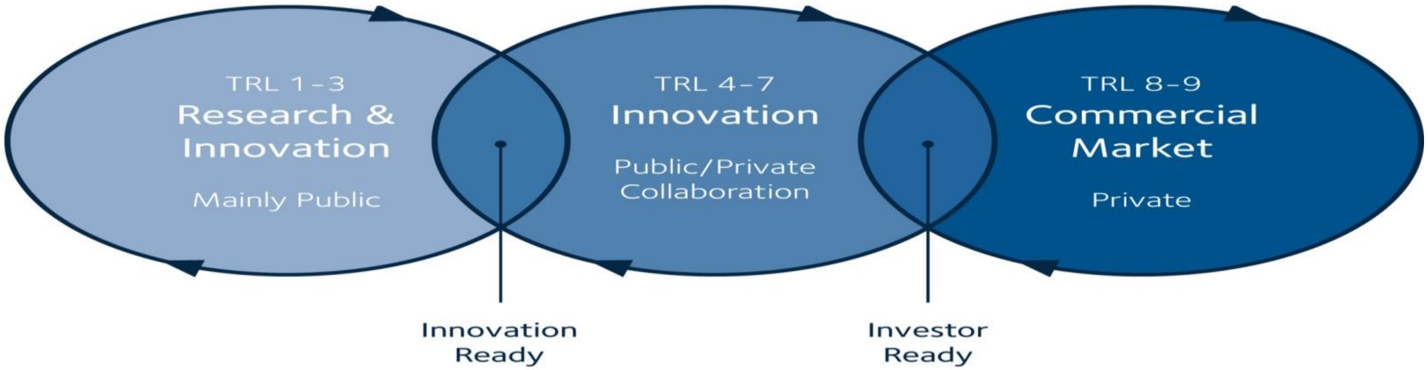
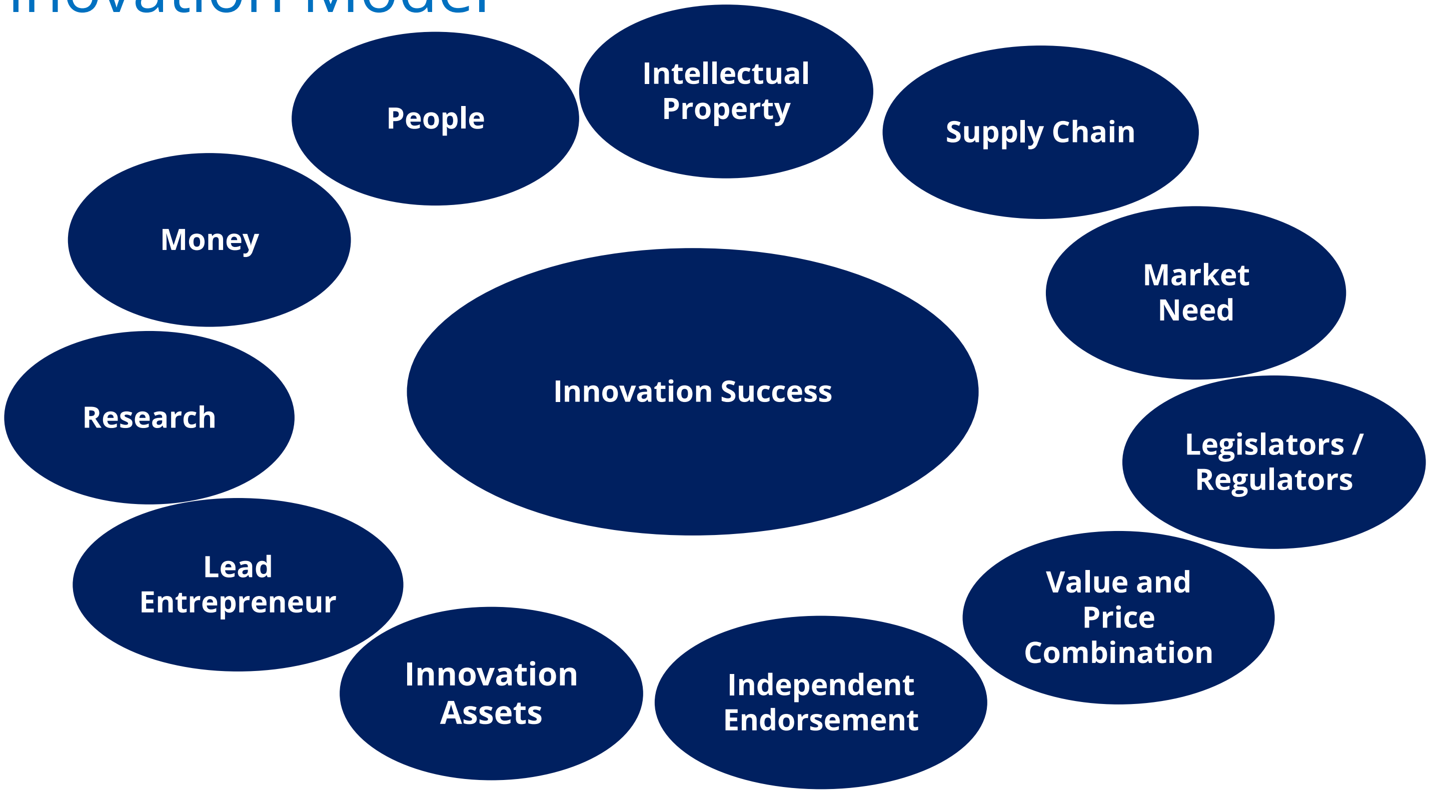


The CPI Innovation Integrator^{TM} Process translates **research** into the clarity of a commercial **business proposition**

CPI Innovation Model



Without all These Factors in Place Innovation Will Fail

Long term **commitment** and **collaboration** are essential

The CPI Innovation Integrator™ Model

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Makes **assets** and **knowledge** available to partners to reduce innovation risk

Converts inventions and **ideas** into everyday processes, products or services

Brings members of the supply chain together into **collaborative partnerships**

Brings together **partnerships** to cover all of the innovation factors

Uses its capability and knowledge to **assess** technologies and business models to **advise** partners on the potential success of their developments

Has **sufficient foresight** to identify future innovation needs and the capability to **support the development** of next generation processes

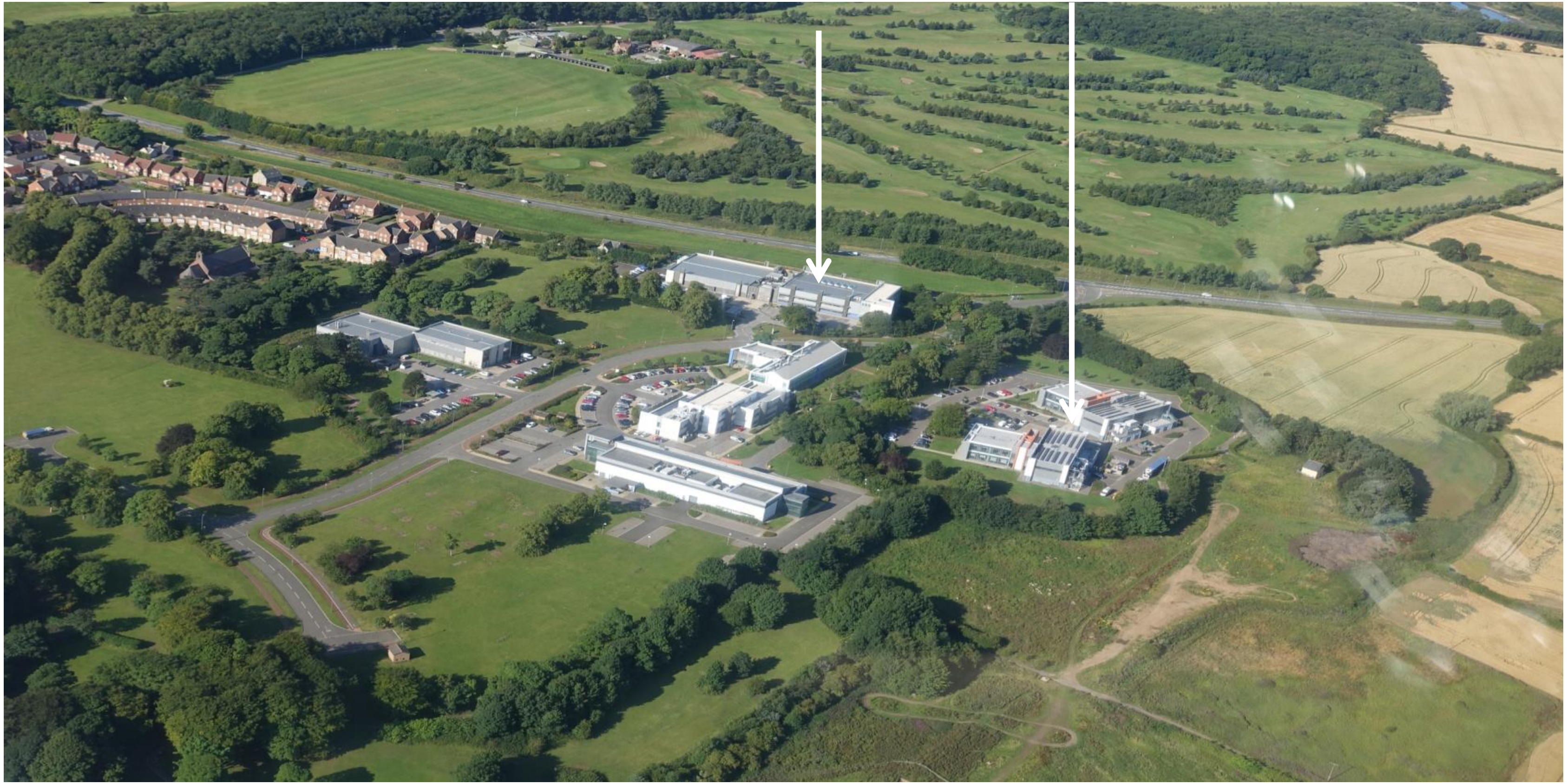
CPI is an innovation integrator that **builds collaborative partnerships** across disciplines and supply chains, and helps those partners to build **value-creating businesses** within the UK. Retaining and developing this model requires CPI to develop **more people** with the **necessary skills**.

How much graphene does your money get you?

Graphene Application Centre - Context

- Why work with a Centre in the UK
  - World class expertise in Printed Electronics and barrier membranes - $50m centre
  - UK National Formulation centre with $45m investment in people and equipment
  - UK National Graphene Applications centre with $20m Investment
  - Industry focused, open access to any company
  - Translation of UK Science with collaborations with: Manchester University NGI, Cambridge Graphene Centre and Durham University

New Facility in Discovery 2 at DCC NETpark



Graphene and nano formulation Centre

Printable Electronics at CPI

World class, **open-access capability** in the scale-up and commercialisation of printable electronic applications

1500m² of clean room facilities

Our Capabilities

Clean rooms (Class 100 and 1000)

Conventional Printing Facilities

Formulation, optical and electrical test laboratory

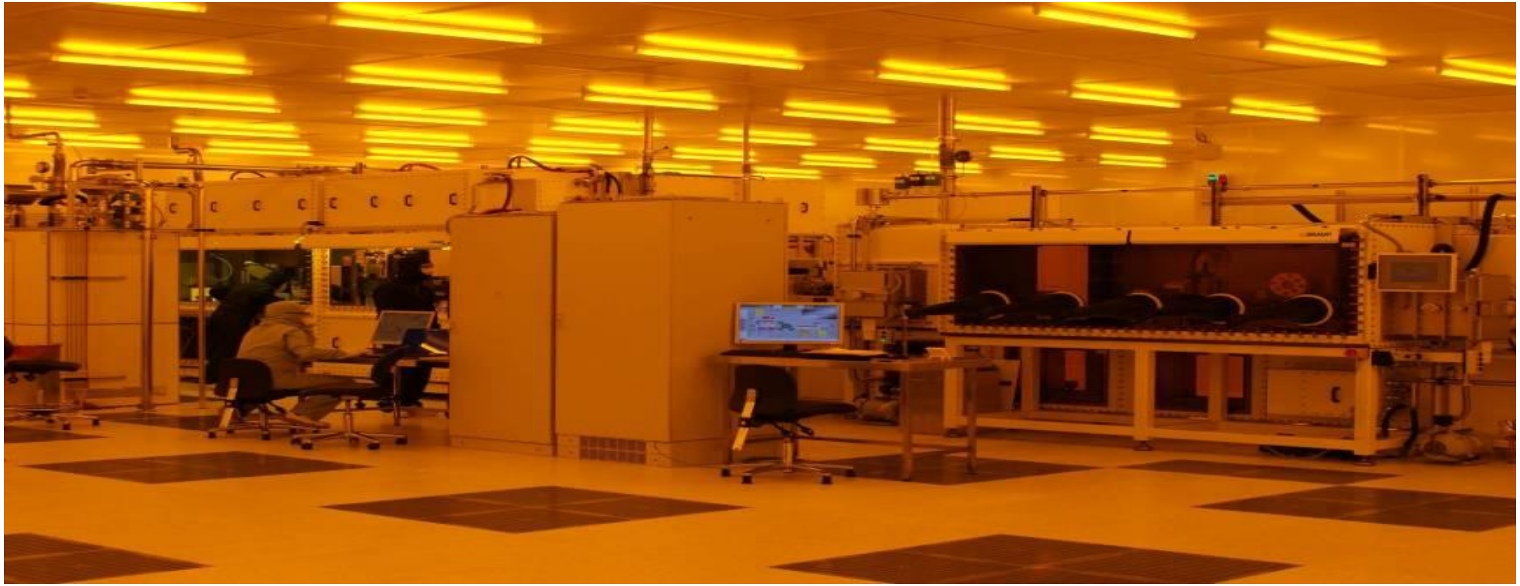
Incubator Office Space

Technical and Commercial teams

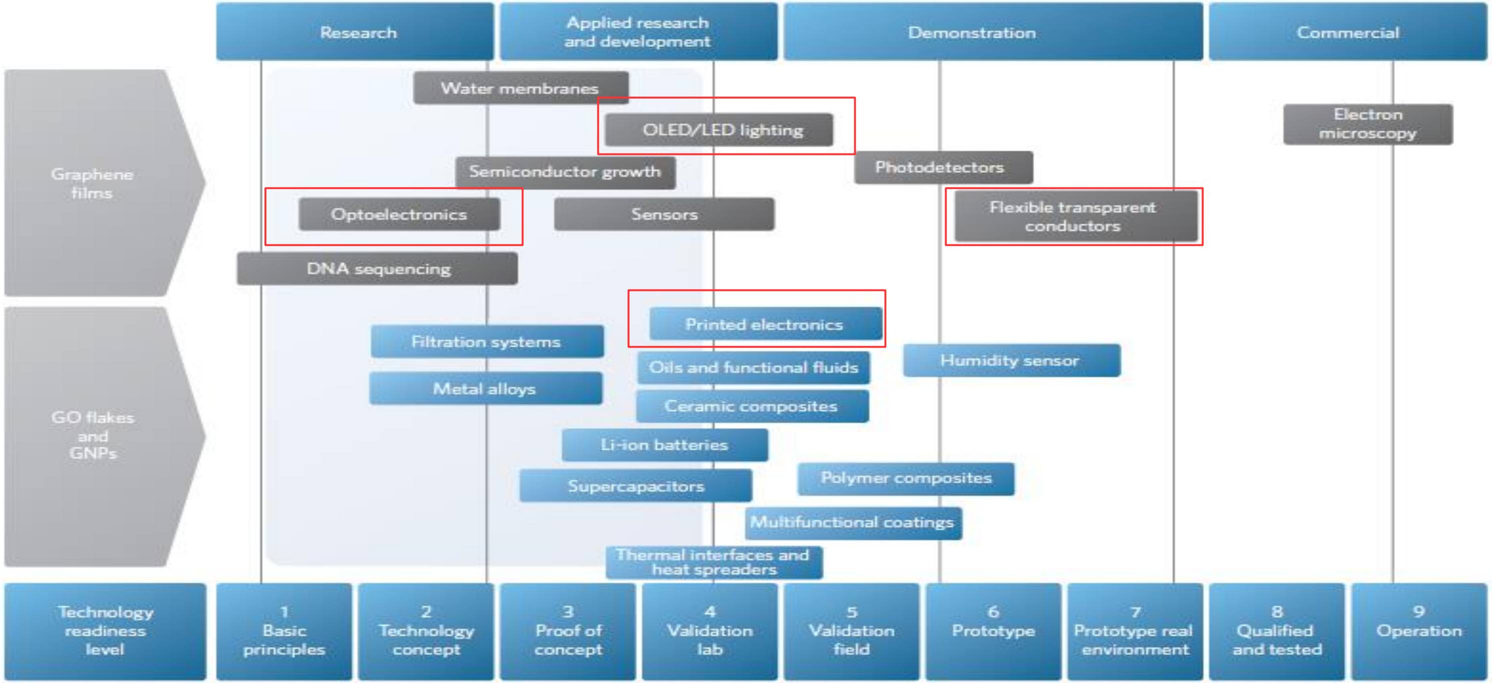
For further information, please see: **www.uk-cpi.com/pe-equipment**

CPI Offering 2: scale and yield projects

Companies proof of manufacture before getting investment



Graphene Applications and Technical level



Graphene applications classified by technology readiness level, NATURE NANOTECHNOLOGY, 2014

The challenges facing the market...

- How to do you know it is graphene?
- How to make graphene consistently ?
- How to measure graphene?
- What are the real benefits of graphene?
- How to handle graphene safely ?

- How to make money from graphene?

What CPI offers

- Open Access
- Collaborative networks
- Awareness Programmes - focused on the dissemination of know-how
- Market Access Assistance
- Training
- Multi-sector market knowledge
- All in one place

The Offers

1) The '*What it is…*' Offer

- An assessment of graphene powder properties
- Quantification of purity and composition
- Surface modification and a measure of the level of surface treatment
- Evaluation of product consistency
- How the above influence fundamental properties of graphene powder

2) The '*What it can do…*' Offer

- Guidelines to formulate graphene
- A view on how formulation affects initial to final properties of dispersed graphene
- Methods to measure electrical, thermal, mechanical, lubrication and optical properties for coatings, inks and composite materials

Method for formulating graphene – Key Steps

Where Offer 1 (C1) & Offer 2 (C2) fit into the process

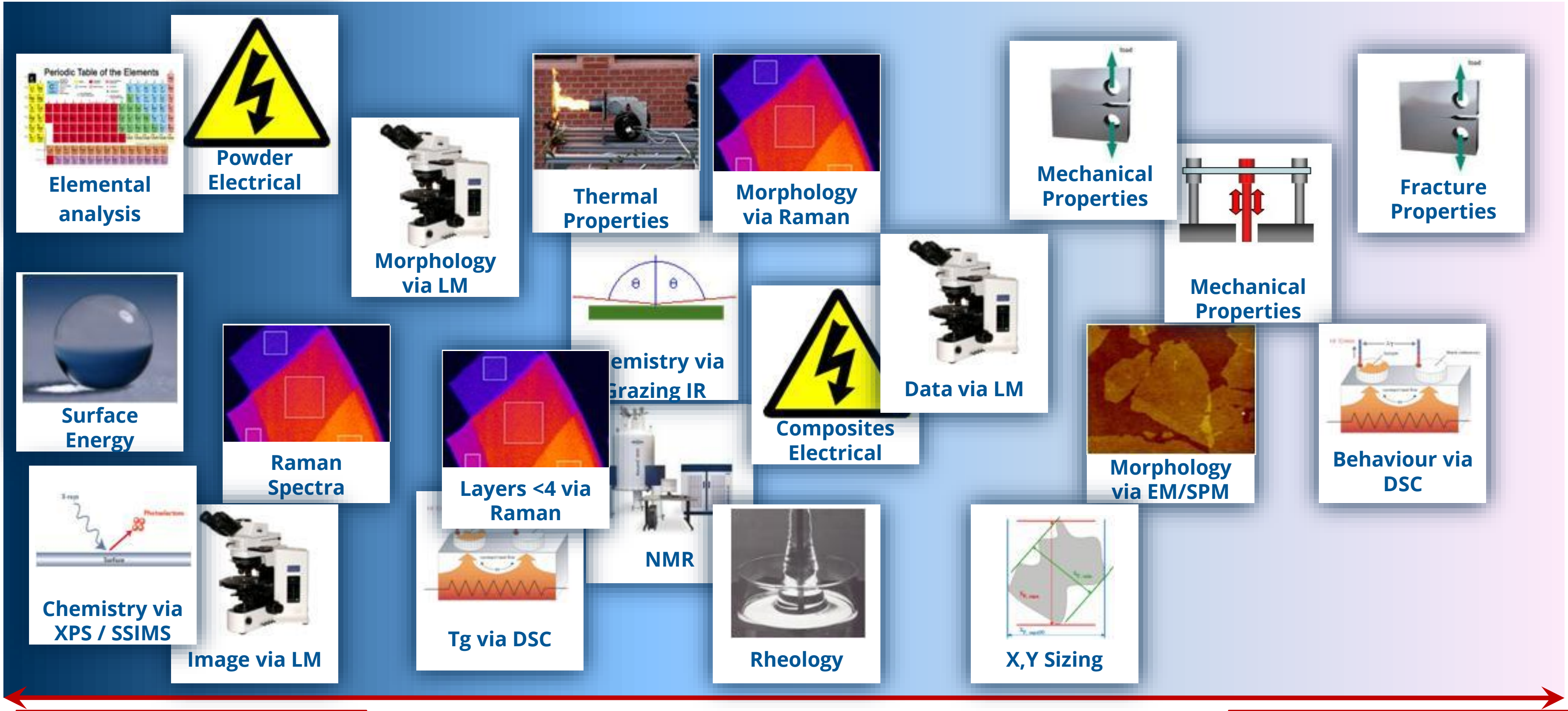
Know your target (C1/C2)
- Target carrier
- Curing / drying method
- Target format (e.g. Ink on PET)
- Any other required additives (e.g. rheology modifiers)
- The properties of interest

Formulate (C1/C2)
- Characterise your graphene
- Match source graphene to the needs
- Purify/disperse/blend
- Build up into the full formulation

Apply and process (C2)
- Apply into the required test geometry

Test (C2)
- Characterise the morphology
- Characterise the properties.

Characterisation – Realising the Offer



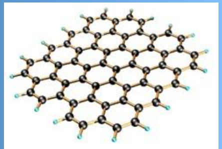
Most Routine

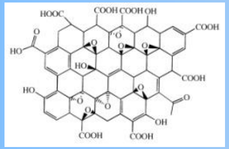
Most Bespoke

Preparation / Compatibility Tools – Realising the Offer

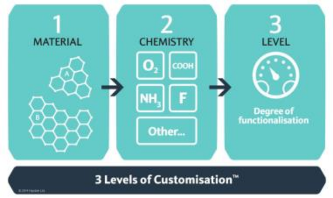
Compatibilisation

G/GO

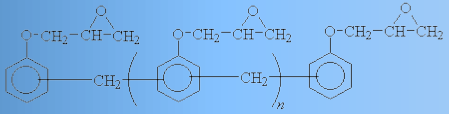




Functionalisation



Matrix



Dispersion

Deposition and Encapsulation

Roll to Roll Deposition Processes

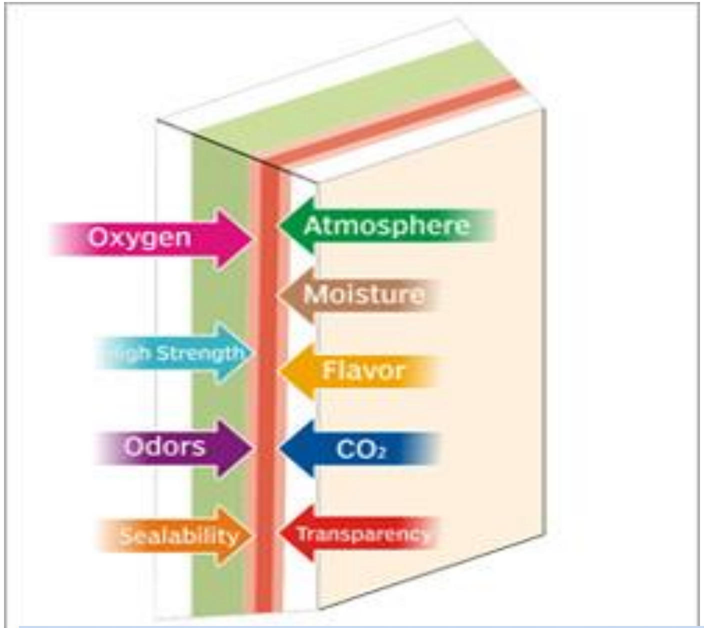
Sheet Deposition Processes

Analytical and Test Equipment

Selection of CPI Projects

- Graphene Barrier – ALD onto Graphene substrates
  - Partners, Prof Hofman, University of Cambridge, Flexenable
  - ALD seal defects in CVD graphene for ultra thin flexible barrier
  - CPI world class barrier expertise, R2R ALD barrier water permeability $10^{\wedge-7}$g/m2/day,
- Water filter
  - Partners, National Nuclear, G20, water filter for nuclear industry
  - Membrane expertise,
- Composites for aerospace
  - Aerospace alliance, Thomas Swan
  - Combination of strength enhancement and removal of copper for lightning strike
- Replacement of carbon black in composites – strength and low cost conductance

Case Study Barrier film - Why do we need Barriers

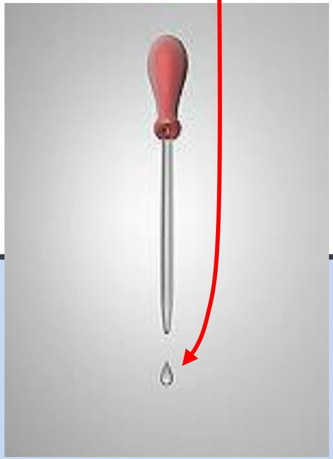


- Organic materials are susceptible to degradation by water and oxygen
- Typical plastic substrates are leaky, letting between 5-20g H_2O/m^2/day permeate through
- Device lifetime is critically determined by efficacy of the barrier and robustness of the chemicals used
- Without a decent barrier lifetimes measured in 10's of hours not 10's of 000's of hours as required by manufacturers and consumers
- Other benefits from Better barrier film:
- Transparency, (remove Aluminium coating)
- Thinner, more flexible and lighter weight films

Water Barrier measurement

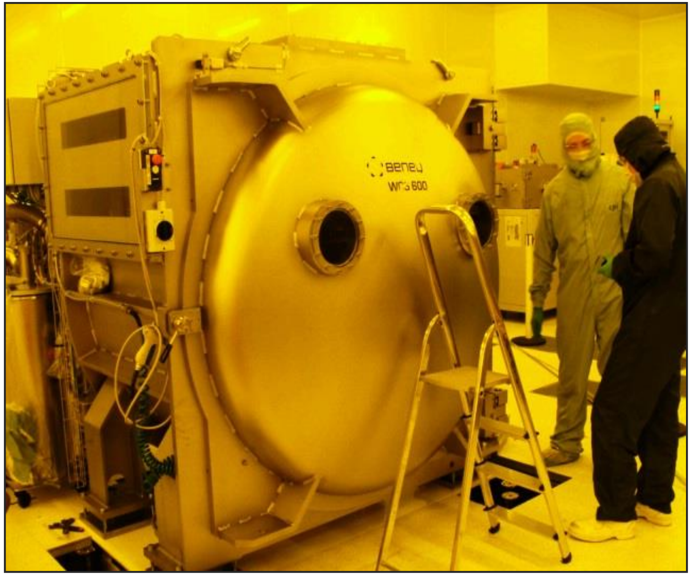
Raw Film | Food Packaging | OPV | OLED

100 | 10 | 1 | 1×10^{-2} | 1×10^{-4} | 1×10^{-6}



R2R ALD Coating for fast high barriers

1st scale-up commercial R2R ALD Process



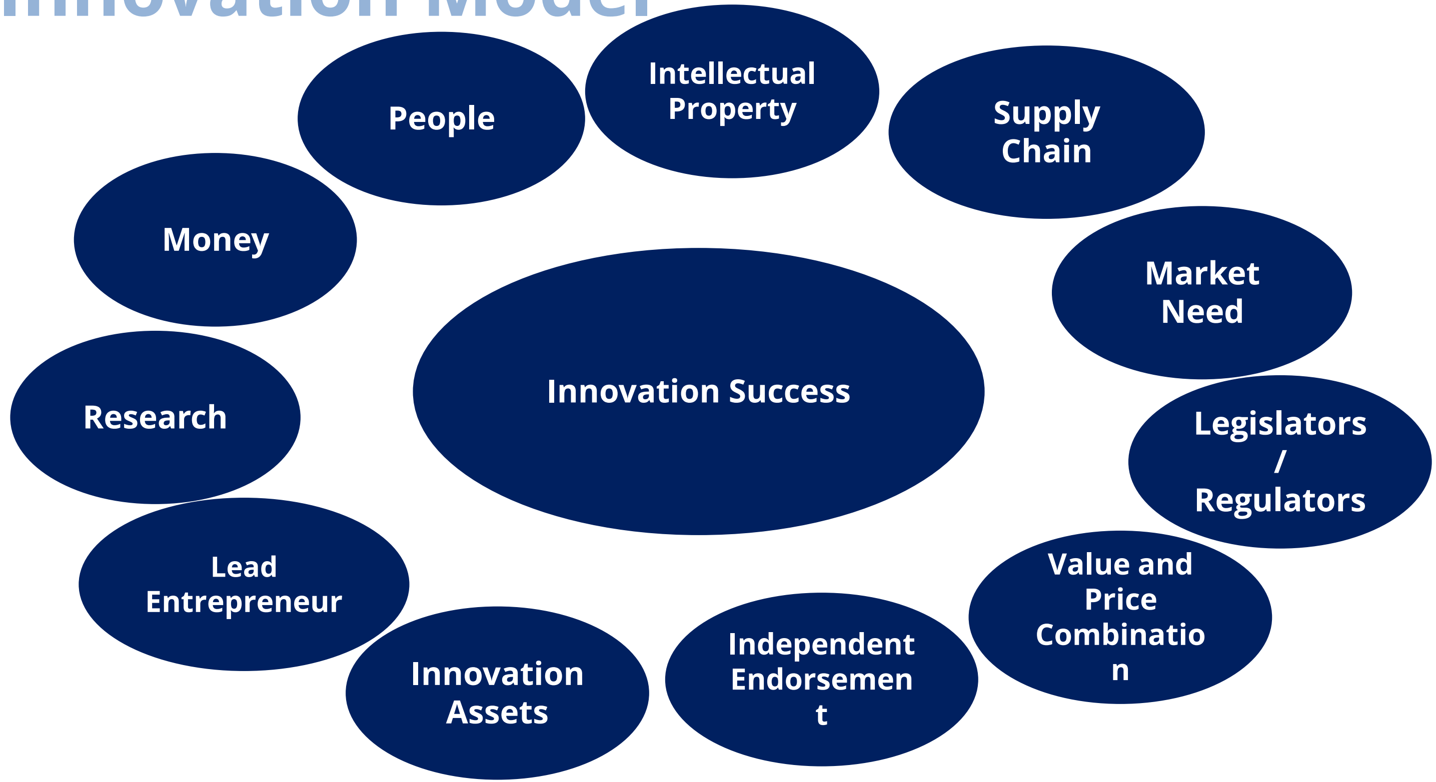
Beneq WCS 600 R2R ALD tool

- 500mm R2R continuous ALD coating tool operation (targeting 1 – 10 m/min)
- Thermal ALD capability with potential for plasma enhancement
- Main precursors TMA
- Conformal self limiting coating technology delivering highly impermeable films 10^{-7}

What we want to avoid

- Repeating the mistakes of previous wonder materials (nano-tubes)
- Unrealistic expectations
- Repetition of fundamental research
- Using poor quality raw materials
- Unfocused use of resources & cash
- Getting caught out by regulation
- Not involved in or influencing new standardisation & safety studies
- Loss of EU position vs. US and Far East

CPI Innovation Model



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Thank you...

For more information visit www.uk-cpi.com

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Twitter: @ukCPI